A

SYNOPSIS

ON

WEB USER BEHAVIOUR ANALYSIS

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by

Abhineet Kr Sinha (2302220100009)
Viswajeet Kr Singh (2302220100010)
Sanjeet Kumar (2302220100163)
Amrit Raj (2302220100032)

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Under the Supervision of

Mr. Manik Chandra Panday



ITS Engineering College, Greater Noida 201306

Affiliated to



Dr. APJ Abdul Kalam Technical University, Lucknow
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ABSTRACT

The rapid growth of digital platforms has increased the need to understand how users interact with websites. Every user action, such as page visits, clicks, navigation paths, and session duration, provides valuable information that can help improve website performance and user experience. However, many existing analytics tools cither provide only basic statistics or arc expensive and difficult for small organizations to customize.

Web User Behaviour Analysis is a web-based application developed to collect, process, analyze, and visualize user interaction data using data analytics and machine learning techniques. The system records user activities and stores them in a database for further processing. Data preprocessing is performed using Pandas and NumPy, while Scikit-learn is used to identify user behaviour patterns through machine learning algorithms. The results are presented through an interactive Streamlit dashboard using Matplotlib, Seaborn, and Plotly for graphical visualization.

The application is developed using HTML, CSS, JavaScript. Bootstrap, and Python Flask, providing a responsive and user-friendly interface, with MySQL/MongoDB used for data storage. The processed information helps website administrators understand visitor behaviour, identify popular pages, improve navigation, and make informed decisions based on user interaction data.

The expected outcome of the project is a simple, intelligent, and cost-effective analytics platform that enables organizations to better understand their website users, improve user engagement, and enhance website performance through data-driven insights.

Keywords: *Web Analytics, User Behaviour Analysis, Machine Learning, Data Visualization, Flask, Streamlit, Web Usage Mining.*

1.Introduction

Understanding how visitors behave once they arrive at a website has become one of the most important challenges for modern organizations. As digital services expand across every sector, the ability to translate raw' interaction data into meaningful, actionable insight has moved from being a competitive advantage to a basic operational necessity.

1.1 Background

The rapid growth of the internet has transformed the way people access information, communicate, shop, and use online services. Websites have become an essential part of businesses, educational institutions, healthcare organizations, banking systems, and e-commerce platforms. Every interaction performed by a user, such as visiting a page, clicking a button, searching for information, or completing a transaction, generates valuable data that reflects user behaviour.

Traditional web analytics tools provide information such as page views, visitor count, and session duration. While these statistics are useful, they do not provide detailed insights into user behaviour, browsing patterns, or user preferences. Data analysis and machine learning techniques can help discover meaningful patterns from user activity, allowing website owners to improve website design, user experience, and decision-making.

1.2 Motivation

Website owners need more than simple visitor statistics to understand how users interact with their websites. Identifying browsing patterns, popular pages, and user engagement helps improve website performance and customer satisfaction. Many available analytics solutions arc expensive or provide limited customization for small organizations and educational projects.

Web User Behaviour Analysis is developed to provide a simple and intelligent analytics platform where website administrators can:

- Collect user interaction data from websites.
- Analyze browsing behaviour using machine learning.
- Visualize user activity through interactive dashboards.
- Monitor website performance.
- Support better decision-making using data-driven insights.

1.3 Proposed Solution

Web User Behaviour Analysis provides an intelligent platform for collecting, processing, and analyzing website user activity. The system records user interactions and applies data processing and machine learning techniques to identify behaviour patterns and generate useful insights. The platform includes the following key features:

- User activity tracking and clickstrcam analysis
- Session analysis and data preprocessing
- User behaviour clustering using machine learning
- Interactive dashboard with graphical data visualization
- User behaviour reports and website performance monitoring

The system enables website owners to understand visitor behaviour, improve website usability, and optimize overall website performance.

1.4 Expected Benefits

The proposed Web User Behaviour Analysis system is expected to provide the following benefits:

- Improved understanding of user browsing behaviour through detailed analysis.
- Better website performance by identifying frequently visited and less engaging pages.
- Interactive dashboards for clear visualization of user activity.
- Data-driven decision making using machine learning insights.
- Enhanced user experience by understanding visitor preferences.
- Cost-effective web analytics solution using open-source technologies.

Collectively, these benefits help organizations improve website usability, increase user engagement, and make informed business decisions based on user behaviour data.

2. Literature Survey

A structured review of existing literature and analytics platforms was carried out to understand the techniques currently used for studying user behaviour on the web, and to identify the gaps that the proposed system aims to address.

2.1 Comparative Analysis of Existing Approaches

A review of existing web analytics platforms and research studies was conducted to understand current techniques for analysing user behaviour. Table 1 highlights the strengths and limitations of commonly used approaches and how the proposed system improves upon them.

Table 1: Comparative Analysis of Literature & Research Approaches

Research / Study	Key Focus & Strengths	Limitations
Srivastava et al. (2000) (Web Usage Mining Frameworks)	Establishes foundational methodologies for discovering access patterns and navigation paths from web logs.	Focuses heavily on theoretical data mining models without providing real-time dashboarding or modern web integrations.
Mobasher et al. (2000) (Automatic Web Personalization)	Uses usage mining to create personalized user experiences and dynamic content recommendation systems.	Involves high computational complexity and lacks flexible customization for general-purpose website performance monitoring.
Kosala & Blockeel (2000) (Web Mining Taxonomy)	Provides a comprehensive survey categorizing Web Content, Structure, and Usage Mining techniques.	Functions as a conceptual survey paper rather than a practical, end-to-end deployable analytical application.
Pedregosa et al. (2011) (Scikit-learn Machine Learning)	Demonstrates efficient Python-based algorithms for data preprocessing, clustering, and behavioral modeling.	Provides general ML libraries without a dedicated web tracking pipeline or user interface for web analytics.

2.2 Existing Research

Several researchers have explored the use of Web Usage Mining and Machine Learning to understand user behaviour [1J. [9], [10]. Studies show that analyzing user interactions such as page visits, clicks, and navigation paths helps improve website usability, personalize content, and enhance customer satisfaction. Machine learning techniques such as clustering and classification arc widely used to group users with similar browsing patterns and predict future user behaviour [2],

Modern visualization tools have also improved the presentation of analytical results by converting complex datasets into easy-to-understand charts and dashboards [3], [4], [6|. These technologies help organizations make faster and more informed decisions based on user activity.

Framework-level studies further show that lightweight, open-source web stacks such as flask [8] combined with data-oriented dashboard frameworks such as Streamlit [7] allow small teams to build and deploy analytical applications quickly, without the licensing overhead associated with commercial analytics suites. This makes such frameworks particularly suitable for academic and small-organization projects.

2.3 Research Gap

Most existing web analytics systems focus on traffic statistics and predefined reports. Few provide an integrated solution that combines data collection, preprocessing, machine learning, and interactive visualization in a simple and cost-effective application.

The proposed Web User Behaviour Analysis system addresses this gap by integrating web technologies, data analytics, machine learning, and visualization into a single platform. It provides meaningful insights that help website owners improve user experience, optimize website performance, and make data-driven decisions.

3. Problem Formulation

3.1 Existing Challenges

With the increasing use of websites for business, education, healthcare, banking, and e-commcrcc, understanding user behaviour has become essential for improving website performance and user satisfaction. Most websites generate large amounts of user interaction data, such as page visits, clicks, navigation paths, and session duration. However, many organizations rely only on basic web analytics tools that provide limited information like visitor count, page views, and bounce rate.

These traditional analytics tools often fail to explain why users leave a website, which pages attract the most attention, or how users navigate through different sections of a website. As a result, website owners find it difficult to identify user interests, improve website design, and make informed decisions. Advanced analytics solutions arc often expensive and require technical expertise, making them less suitable for small businesses, startups, and educational organizations.

3.2 Proposed Solution

The proposed Web User Behaviour Analysis system addresses these challenges by providing a simple and intelligent web analytics platform. The system collects user interaction data, stores it in a database, and processes it using data analysis and machine learning techniques.

The application analyzes user behaviour by identifying browsing patterns, user engagement, and navigation trends. It presents the results through an interactive dashboard with graphical visualizations, allowing website administrators to understand visitor behaviour and improve website performance.

The system is developed using HTML, CSS, JavaScript, Bootstrap, Python Flask, MySQL/MongoDB, Pandas, NumPy, Scikit-learn, Streamlit, Matplotlib, Seaborn, and Plotly. By combining web technologies with machine learning and data visualization, the proposed solution provides a cost-effective and user-friendly platform for analysing website user behaviour and supporting data-driven decision making.

4.Objectives

To achieve the core vision of this project, the proposed work is structured around the following key objectives**:**

- To study existing web analytics techniques and user behaviour analysis methods through a detailed literature survey.
- To develop a web-based application that collects, processes, and analyses website user activity using data analysis and machine learning techniques.
- To design an interactive dashboard that visualizes user behaviour patterns and helps website administrators improve user experience and website performance.
- To evaluate the performance of the proposed recommendation system using appropriate error metrics (e.g., accuracy, precision, recall, and F1-score / MSE).

5.Methodology / Planning of Work

The proposed system will be built incrementally, moving from requirement gathering and design in the early part of the project to full-stack development, machine learning integration, and deployment in the later stages. This phased approach ensures that each component is validated before the next is built on top of it, reducing rework and keeping the project aligned with the defined objectives.

The development of the proposed system will be carried out in the following phases, planned across the seventh and eighth semesters:

Table 2: Phase-Wise Methodology of the Proposed Work

Phase	Activities
Phase 1 - Requirement Analysis	Study existing web analytics systems, identify project requirements, and define user behaviour metrics.
Phase 2 - System Design	Design the system architecture, database schema, and user interface
Phase 3 - Development	Develop the frontend using 11TML, CSS, JavaScript, and Bootstrap, and the backend using Python Flask.
Phase 4 - Data Processing & Machine Learning	Store user activity data, preprocess it using Pandas and NumPy, and apply machine learning algorithms using Scikit-learn.
Phase 5 - Dashboard & Testing	Develop the Streamlit dashboard, create visualizations, and perform functional and usability testing.
Phase 6 - Deployment	Deploy the application on Render or Heroku and prepare the final documentation.

6. Activity Chart

The project activities arc planned across the seventh and eighth semesters to ensure systematic development and timely completion, as shown in Table 3.

Table 3: Semester-Wise Activity Chart

Activity	Semester VII	Semester VIII
Project Topic Selection	✓	
Literature Survey	✓	
Requirement Analysis	✓	
System Design	✓	
Database Design	✓	
Frontend Development	✓	✓
Backend Development	✓	✓
Database Integration	✓	✓
Data Collection & Preprocessing		✓
Machine Learning Model Development		✓
Dashboard Development		✓
Data Visualization		✓
Testing & Bug Fixing		✓
Deployment		✓
Documentation & Final Report		✓
Project Presentation & Viva		✓

7. SDG Goals Relevance

Table 4 presents the mapping of the proposed project to the relevant Sustainable Development Goals (SDGs).

Table 4: Mapping of the Project to Sustainable Development Goals

Project Name	Problem Description	SDG No.: SDG Name
Web User Behaviour Analysis	Many organizations struggle to understand user interaction and improve website performance using traditional analytics tools. The proposed system uses machine learning and data visualization to provide meaningful insights into user behaviour and support better decision-making.	SDG 9: Industry, Innovation and infrastructure SDG 8: Decent Work and Economic Growth

Justification

- The project promotes digital innovation by applying machine learning and data analytics to improve website performance and user experience (SDG 9).
- 11 provides an affordable and intelligent analytics solution that helps businesses make data-driven decisions and improve digital services (SDG 9).
- By helping organizations optimize their online platforms and increase user engagement, the project supports productivity, business growth, and innovation (SDG 8).

8. Type of Project

The proposed work, Web User Behaviour Analysis, is an Application-Based Project with a supporting Research component.

- Application-Based: The project delivers a fully functional, deployable web application that collects, processes, and visualizes real user interaction data through a working Flask-based backend, a responsive frontend, and an interactive Streamlit dashboard.
- Research-Oriented: The project involves the study and application of machine learning techniques for behaviour pattern discovery, drawing on Web Usage Mining literature to guide the design of the analytical component.

This combination ensures that the project is grounded in established research while producing a practical, usable analytics platform that addresses a real-world need for organizations seeking to understand their website visitors.

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